

Assylkhan Geniyat

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Research Interests: Large Language Models, Retrieval-Augmented Generation, Agentic AI, Reasoning, Efficient AI Systems

Education

Boston University, Graduate School of Arts & Sciences

Master of Science in Artificial Intelligence

Boston, MA, USA

Jan. 2026 – May 2027 (Expected)

- **GPA: 4.00 / 4.00**

Astana IT University

Bachelor of Science in Computer Science

Astana, Kazakhstan

Sep. 2022 – Jul. 2025

- **GPA: 3.65 / 4.00**; graduated with Honors.
- Bachelor's thesis: *AI for Drug Discovery Using Molecular Docking*.

Research Experience

First-Author Research – Retrieval-Augmented Generation

2026

Boston University, Graduate School of Arts & Sciences

- Led a first-author RAG study using a **608-question TechQA dataset** spanning **496 source documents**, with separate training, validation, and 208-question reserve partitions.
- Designed controlled experiments comparing hybrid retrieval, an unadapted neural reranker, document-supervised adaptation, and evidence-specific supervision across **4 chunking strategies**.
- Conducted external-transfer evaluation on **273 Doc2Dial documents** grouped into **235 dependency families**, using **3 paired training seeds** to evaluate generalization beyond the original domain.
- Implemented end-to-end RAG pipelines using BM25 and dense retrieval, reciprocal-rank fusion, Qwen3 neural reranking, LoRA adaptation, and LLM generation with **Qwen3.5** and **Mistral 7B** across project iterations.
- Executed and analyzed **8,736 generation records** and **152,880 reranker candidate scores** in the external-transfer study.
- Conducted uncertainty analysis using **20,000 bootstrap resamples** and **100,000 paired sign-flip tests**, together with answer-level and retrieval-quality evaluation.
- Led conceptualization, methodology, software development, experimentation, formal analysis, visualization, project administration, and manuscript preparation.

Research Assistant

Dec. 2023 – Jul. 2024

Astana IT University

Supervisor: Ida Akhmetvaliyeva

- Conducted research on computational methods for analyzing and generating effective Russian-language media headlines.
- Contributed to experimental design, data analysis, evaluation, and manuscript preparation.
- Co-authored a peer-reviewed IEEE publication.
- Presented research findings and received **2nd place among 27 AI research papers**.

Publications & Manuscripts

1. **A. Geniyat**, Z. Fakhriddinov, A. Gulati. “Does Evidence-Specific Supervision Improve Support-Answer Generation? A Controlled Study of Literal-Reference Supervision and External Transfer.” *Manuscript under review*, Engineering Applications of Artificial Intelligence (EAAI), 2026.
2. Z. Fakhriddinov, **A. Geniyat**, A. Gulati. “Answer Support Is Not Citation Correctness: A Matched-Evidence Study of Local Scientific RAG.” *Manuscript under review*, Engineering Applications of Artificial Intelligence (EAAI), 2026.
3. D. Alkhaidar, **A. Geniyat**, I. Akhmetvaliyeva. “The Role of Computer Technologies in Formulating Effective Headlines for Russian-Language Media.” *IEEE: Digital Generation*, 2024, pp. 140–145. [DOI](#)

Professional Experience

Instructional Technology Developer (Data & Automation)

Jan. 2026 – Present

Boston University

Boston, MA

- Develop AI agents and automation workflows supporting instructional technology, university operations, and data-driven decision making.
- Contributed to an **AI-powered faculty analytics platform** designed to support teaching evaluation, feedback analysis, and instructional improvement at university scale, with potential applicability across Boston University's **4,500+ faculty members**.

- Build API- and webhook-based integrations connecting learning management systems, analytics platforms, databases, and internal university tools.
- Develop automated data pipelines and reporting workflows supporting faculty and administrative decision making.

Machine Learning Engineer (Primary AI/ML Engineer)

Jul. 2025 – Dec. 2025

National Bank of Kazakhstan

Astana, Kazakhstan

- Served as the **primary AI/ML engineer** responsible for end-to-end design, development, deployment, and technical ownership of **five production AI/ML systems**.
- Built and deployed an enterprise AI-agent and retrieval system serving **6+ internal departments**, enabling semantic search and document question answering across more than **80,000 financial and legal documents**.
- Designed the end-to-end RAG architecture, including retrieval, embeddings, metadata-aware search, reranking, vector search, and LLM-based answer generation.
- Led AI projects from problem formulation and data preparation through model development, system integration, deployment, and production use.

Machine Learning Engineer

Mar. 2025 – May 2025

Situational-Analytical Center of the Fuel and Energy Complex

Astana, Kazakhstan

- Developed a multilingual NLP system processing approximately **25,000 news articles per day** in three languages, achieving **97.2% classification accuracy**.
- Reduced manual news-labeling workload by approximately **95%** through automated classification.
- Developed classification systems for **five government ministries**, improving average recall from **78% to 90%**.

Honors, Scholarships & Selected Training

- **Bolashak International Scholarship**, Government of Kazakhstan – fully funded government scholarship supporting graduate study at Boston University.
- **Boston University Merit Scholarship** – \$20,000.
- **2nd Place, AI Research Paper Competition** – selected among 27 AI research papers, 2024.
- **Full-Tuition Merit Scholarship**, Astana IT University.
- **Samsung Innovation Campus Microqualification** – GPA: **3.91 / 4.00**; final project evaluated as **Excellent**.

Academic Leadership & Service

Conference Organizer, 2024 IEEE AITU: Digital Generation

Sep. 2023 – Apr. 2024

Astana IT University

- Served on the organizing team of the **first IEEE student conference held in Kazakhstan**, leading IT operations, developing the official conference website, and providing technical support.
- Helped deliver an international research conference featuring **200+ research papers from 45 universities across 7 countries**.

Vice President, SIAM Student Chapter

2022 – 2025

Astana IT University

- Organized a student meeting with representatives from **Boston University** and **Huawei**.
- Mentored **400+ first-year students** in Linear Algebra through peer-led academic support sessions.
- Helped organize seminars, workshops, and academic activities in applied mathematics and computer science.